CTF-GS: Compact Teacher Feature Fields with View-to-Primitive Registration for Open-Vocabulary 3D Gaussian Scene Understanding

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Abstract

We study whether dense vision-foundation features can be reconstructed as a 3D Gaussian scene representation and reused at novel views for open-vocabulary scene understanding. Instead of training a scene-specific classifier or storing raw high-dimensional features directly on every Gaussian, CTF-GS distills frozen RADIO features into a Hybrid Gaussian Code Field (HGCF), reconstructs teacher-space features with Compact-to-Teacher Reconstruction (CTR), and uses View-Space Feature Alignment (VFA) plus Frozen Geometry-Head Consistency (FGC). We further expose feature quality and visibility as explicit multi-head field outputs supervised by label-free teacher agreement and rendered alpha. The resulting scene representation supports rendered dense feature maps for novel-view grounding and direct Gaussian-center primitive scores for 3D queries. A label-free VPR-to-field consistency stage can also transfer registered view evidence back into the compact direct field. On LERF-OVS, the current frozen RADIO-GS package with a GT-free peak-component coarse mask and feature-only SAM3 boundary readout reaches 0.8598 macro localization accuracy and 0.5889 macro mIoU across four scenes. On the VALA/OpenGaFF eight-scene ScanNet direct point-query protocol, the promoted DINO-CV contextual kNN point readout with label-free scene-mean calibration and spatial logit propagation reaches 0.3806, 0.3871, and 0.4711 mIoU on the 19-, 15-, and 10-class splits. An OpenGaussian-style direct 3D object-selection evaluation shows that a compact prompt-ensemble readout with a GT-free RGB/score-component support guard reaches 0.5014 macro mIoU and 0.7044 Acc@0.25, without reading a VPR cache or using official RGB SAM at inference. The strict no-RGB one-map ablation reaches 0.4570 macro mIoU and 0.6851 Acc@0.25. With a frozen official SAM3 box-prompt diagnostic readout after compact 3D selection, the field reaches 0.5705 macro mIoU and 0.6835 Acc@0.25 under a fixed global threshold. For context, the raw Gaussian-center diagnostic reaches only 0.012/0.009 under the fixed top-2% selector used in the VPR sweep and 0.0804/0.0932 under an earlier top-10% selector; under the same threshold-sweep diagnostic protocol, confidence-weighted VPR-to-field consistency lifts the direct Gaussian-center readout from 0.0458/0.0707 to 0.4363/0.6191. These results support the central claim that 3D Gaussian scenes can serve as reusable foundation-feature memories across rendered-view and registered primitive-level readouts, while clarifying that raw Gaussian-center readout is weak and Waldo Kitchen remains a difficult object-fragmentation case.

1. Introduction

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding, this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views.

CTF-GS targets this gap by reconstructing teacher foundation features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether a scene representation can support two complementary interfaces from the same compact field: dense novel-view feature maps for rendered open-vocabulary localization, and registered primitive/point-level features for direct 3D selection or semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Hybrid Gaussian Code Field, decode it through a Compact-to-Teacher Reconstruction module, and refine it in view space before comparison to the frozen RADIO teacher. We also use Frozen Geometry-Head Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

Our current paper package is conservative. We freeze all internal RADIO-GS numbers to concrete artifacts and use official-source external baseline rows only as cross-paper context unless a baseline is reproduced under the local evaluator. The result is a clear submission route: LERF-OVS provides the main open-vocabulary grounding evidence, the VALA/OpenGaFF-8 ScanNet direct point-query probe tests cross-domain feature usability, and profiled evaluation runs quantify runtime and memory.

Contributions. This draft makes four claims:

- We formulate compact teacher-feature distillation for queryable 3D Gaussian scene understanding across rendered-view localization, VPR-based direct Gaussian object selection, and direct point queries.
- We introduce a unified multi-head Hybrid Gaussian Code Field with compact latent storage, a coarse spatial branch, and explicit feature-quality and visibility readouts, avoiding direct high-dimensional teacher-feature storage on every Gaussian.
- We use Compact-to-Teacher Reconstruction, View-Space Feature Alignment, and Frozen Geometry-Head Consistency to recover teacher-compatible features while keeping teachers and downstream heads fixed.
- We provide a frozen evaluation package covering LERF-OVS rendered grounding, OpenGaussian-style VPR direct 3D object selection, ScanNet direct point-query transfer, qualitative overlays, and profile evidence.

2. Related Work

Language fields and open-vocabulary 3D understanding. LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [5]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [12, 14]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [15]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [6], while InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [7, 11]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [3], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [9]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [8]. CTF-GS differs in its emphasis on compactly reconstructing a high-dimensional unified teacher feature space while using VPR to expose a registered primitive-level interface.

Gaussian scene representations. 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [4]. CTF-GS keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

Agglomerative vision foundation models. RADIO-style models distill multiple visual capabilities into a single dense vision backbone [10, 13]. CTF-GS uses C-RADIOv4-H as a frozen teacher and studies whether its dense features can be reconstructed from 3D scenes at novel views.

Foundation-model regularization in 3DGS. FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [16]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors into 3D Gaussian features for promptable object segmentation [1]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

3. Method

3.1. Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a feature renderer that maps a novel camera view to a dense feature map compatible with a frozen RADIO teacher. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The teacher target is:

$$F_v^\star = T(I_v) \in \mathbb{R}^{H\times W\times 1280}. \tag{1}$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against $F_v^\star$.

3.2. Hybrid Gaussian Code Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. CTF-GS attaches compact learnable latent features to the Gaussians and combines two paths:

1. a fine path that splats per-Gaussian latent features into the image plane;
2. a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused by decoupled geometry and semantic heads into a compact feature map Z_v. In the unified multi-head field, the same fused state also predicts a

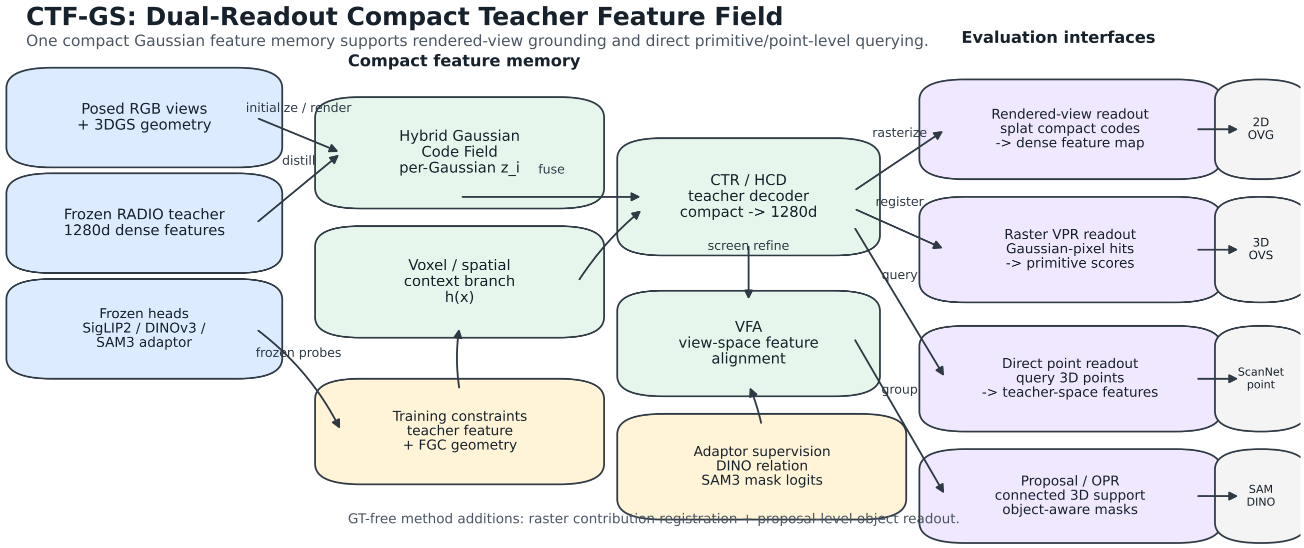

Figure 1. CTF-GS framework. A compact multi-head hybrid Gaussian teacher-feature field stores per-Gaussian codes and spatial context, decodes them into teacher-space features, predicts quality/visibility, and supports three readouts: rendered dense feature maps for novel-view open-vocabulary grounding, VPR primitive scores for direct 3D object selection, and direct point features for ScanNet semantic querying. A frozen official SAM3 box-prompt readout can refine the rendered selection boundary after primitive scoring. Optional raster-contribution and proposal/OPR branches are implemented as audited diagnostics. Frozen RADIO/SigLIP2/DINOv3/SAM3 heads supervise feature, relation, geometry, and mask-logit consistency without training task-specific evaluators; a feature-only prompt-conditioned SAM3 boundary head is additionally trained from official SAM3 training-view pseudo masks and evaluated without RGB SAM3 calls.

feature-quality logit q_v and a visibility logit u_v:

$$(Z_v, q_v, u_v) = H_\phi(Z_v^{fine}, Z_v^{coarse}, d_v, \alpha_v). \quad (2)$$

The quality head estimates whether a rendered compact feature will reconstruct the frozen teacher reliably, while the visibility head estimates whether the rendered evidence is supported by visible Gaussian mass. Both heads are available for rendered-view readout and direct Gaussian/point readout. A Compact-to-Teacher Reconstruction module (implemented as the HCD codec in the codebase) maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(Z_v). \quad (3)$$

A View-Space Feature Aligner (implemented as the screen-space refiner in the codebase) then improves local alignment using rendered guidance signals such as visibility, alpha, and geometric depth.

3.3. Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^\star \rangle}{\|\hat{F}_v\|_2 \|F_v^\star\|_2} + \lambda_{rec}\|\hat{F}_v - F_v^\star\|_1. \quad (4)$$

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc}\mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \quad (5)$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

The unified quality heads add two label-free auxiliary objectives:

$$y_v^q = \frac{1 + \cos(\hat{F}_v, F_v^\star)}{2}, \qquad y_v^u = \alpha_v, \quad (6)$$

where gradients are stopped through the targets. We train q_v and u_v with binary cross-entropy against y_v^q and y_v^u respectively. These losses do not introduce task labels; they expose feature reliability and visibility as first-class field outputs that can be reused by text grounding, direct primitive scoring, and failure analysis.

3.4. Frozen Adaptor Consistency

C-RADIOv4-H exposes frozen adaptor heads for `siglip2-g`, `dino_v3`, and `sam3`. The main open-vocabulary evaluator uses the text-aligned `siglip2-g` path, while training can additionally supervise the rendered feature through the other adaptors. For an adaptor A_a, we add a pointwise consistency term:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^\star)). \tag{7}$$

For dino_v3, we also match the token self-similarity structure rather than only the local feature value. Given normalized adaptor tokens D and $D^\star$, we compute:

$$\mathcal{L}_{rel} = \|DD^\top - D^\star D^{\star\top}\|_2^2. \tag{8}$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. We also evaluate a cross-view variant inspired by ProFuse's use of dense cross-view correspondence and context fusion [2]. For two training views p and q, we match teacher and rendered DINO token affinity:

$$\mathcal{L}_{cv} = \|D_p D_q^\top - D_p^\star D_q^{\star\top}\|_2^2. \tag{9}$$

Unlike ProFuse, this is an optimization-time regularizer on rendered feature maps, not a full pre-registration or context-proposal pipeline. To prevent cross-view smoothing from erasing text-grounding peaks, we add a label-free SigLIP text-heatmap distillation term over a fixed LERF text bank. The per-pixel query variant preserves the scene-category distribution:

$$\mathcal{L}_{txt}^{query} = \mathrm{KL}(\mathrm{softmax}(S^\star T^\top/\tau) \,\|\, \mathrm{softmax}(\hat{S}T^\top/\tau)). \tag{10}$$

We also evaluate a spatial variant that normalizes each query over image locations, directly targeting the LERF argmax localization failure mode:

$$\mathcal{L}_{txt}^{spatial} = \mathrm{KL}(\mathrm{softmax}_{xy}(s_q^\star/\tau) \,\|\, \mathrm{softmax}_{xy}(\hat{s}_q/\tau)). \tag{11}$$

For sam3, the current implementation uses a mask-free soft-region loss. Teacher SAM3 tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by teacher similarity, and the rendered feature is trained to match the teacher region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\mathrm{sam3}}(\hat{F}_v)) - \mu_k(A_{\mathrm{sam3}}(F_v^\star))\|_2^2. \tag{12}$$

This approximates SAM-style region supervision when external SAM mask caches are not available, while keeping the inference-time representation unchanged. We additionally implement a mask-logit distillation fallback for the sam3 adaptor. Teacher SAM3-adaptor tokens define deterministic mask anchors, and we match the rendered feature's per-token soft assignment logits to the teacher distribution:

$$\mathcal{L}_{mask} = \mathrm{KL}\Big(\mathrm{softmax}(A_{\mathrm{sam3}}(F_v^\star)P^\top/\tau) \,\|\, \mathrm{softmax}(A_{\mathrm{sam3}}(\hat{F}_v)P^\top/\tau)\Big) \tag{13}$$

where P are teacher anchor tokens. This remains an adaptor-space training loss rather than an official frozen SAM3 mask-decoder loss. The official SAM3 weights are used only in the separate promptable readout/cache path, where the decoder is frozen and does not train the evaluator.

3.5. Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

3.6. Open-Vocabulary Grounding

At evaluation time, CTF-GS renders a feature map for each novel view. For a text query q, we compute a text embedding e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ. The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center readout decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_\theta(c_i), \qquad s_i(q) = \cos(A_{\mathrm{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This readout is structurally the same compact field used by ScanNet point-query experiments. For LERF primitive selection, we use a frozen SigLIP2 prompt ensemble and a support-aware primitive policy: selected Gaussians are rendered to a coarse mask, RGB/GrabCut snaps local boundaries, and a component guard keeps the largest component only when it dominates the refined support; otherwise it preserves multi-component support only above a fixed support-area floor. This is a GT-free readout rule and uses LERF masks only for the final query-select-render evaluation.

We also retain *View-to-Primitive Registration* (VPR) as a diagnostic and teacher. VPR renders teacher-space features from posed views, applies VFA and the frozen SigLIP2 head, registers those text-aligned features back to visible Gaussian centers with depth/alpha checks, and then runs the same primitive-level text query on the registered Gaussian embeddings. To make the direct compact field itself more usable, we additionally train a label-free VPR-to-field consistency variant. Let $\bar{e}_i^{\mathrm{vpr}}$ be the cached normalized summary feature registered to Gaussian i by VPR, and let n_i be the number of VPR views that registered valid evidence to Gaussian i. The promoted transfer uses a GT-free confidence weight $w_i = \log(1+n_i)/\mathrm{mean}_j \log(1+n_j)$ over valid registered primitives. We fine-tune the compact field and a lightweight summary adaptor with

$$\mathcal{L}_{v2f} = 1 - \cos(A_{\text{SigLIP2}}(D_\theta(c_i)), \text{sg}[\bar{e}_i^{\text{vpr}}]) + \lambda\,\mathcal{L}_{text},$$

where $\mathcal{L}_{text}$ matches the registered VPR scene-query distribution and uses only the LERF text categories, not object masks; all loss terms are averaged with w_i. This stage is a direct-field transfer objective: the VPR-to-field variant tests whether registered multiview evidence can be compressed back into the queryable Gaussian-center field. The paper-facing compact row does not read the VPR cache at inference; the frozen SAM3 box-prompt boundary readout described below is reported as a diagnostic post-selection boundary readout rather than the main compact-only claim. For the Dr. Splat-inspired audit, we also implement rasterizer-level Gaussian–pixel hit registration variants: all footprint contributions, per-pixel dominant hits, and per-Gaussian top-contribution pixels. These variants capture real rasterization intersections instead of only projected centers, but they are reported as diagnostics because they reduce the current LERF direct-3D metrics under the fixed protocol.

4. Experiments

4.1. Submission-Freeze Protocol

The conservative RADIO-GS numbers come from the 2026-05-02 submission-freeze report: `output/radio_gs/reports/submission_freeze_report.md`. The RADIO adaptor ablations were completed on 2026-05-03 under the same LERF sweep protocol. External baseline rows are now official-source values from the cited papers or supplements and are captioned as cross-paper context rather than reproduced same-evaluator results; the exact source rows and protocol caveats are recorded in `baseline_source_verification.md`.

4.2. LERF-OVS Main Result

Table 1 is the current main internal result. The readout first computes the same rendered SigLIP2 heatmap as the localization metric, removes disconnected thresholded regions that do not contain the query peak, and then uses this mask as a coarse prompt for an internal SAM3-style boundary head trained from official SAM3 pseudo masks on unlabelled training views. A refined mask is accepted only if it preserves the query peak and enough normalized heatmap mass, so the module can sharpen boundaries without moving away from the rendered feature response. Relative to the promoted peak-component core readout, macro mIoU rises from 0.5707 to 0.5889 with unchanged LocAcc.

Table 2 provides the paper-facing external comparison. The LERF, LangSplat, and LEGaussians rows are traced to official paper or supplement tables and converted to decimal LocAcc when originally reported as percentages. Because

Table 1. Frozen CTF-GS LERF-OVS rendered-feature grounding result. mIoU uses a GT-free peak-relative mask threshold of 0.60, a peak-connected-component coarse mask, and a feature-only prompt-conditioned SAM3 boundary readout with heatmap-support acceptance. The boundary readout receives rendered CTF-GS features, the text prompt, and the coarse mask; it does not call RGB SAM3 at evaluation time. LocAcc is measured from the heatmap argmax before mask extraction.

Scene	LocAcc	mIoU	Temp.
Figurines	0.8214	0.5243	50
Ramen	0.9014	0.6325	40
Teatime	0.8983	0.6515	25
Waldo Kitchen	0.8182	0.5475	25
Macro	0.8598	0.5889	–

Table 2. LERF-OVS open-vocabulary grounding comparison. External rows are official-source LocAcc values converted to decimals; CTF-GS is the local rendered-feature result. Best per column is in **bold**.

Method	Figurines	Ramen	Teatime	Waldo Kitchen	Macr
LERF	0.795	0.625	**0.938**	0.815	0.79
LangSplat	0.804	0.732	0.881	**0.955**	0.84
LEGaussians	0.767	0.737	0.683	0.523	0.67
RADIO-GS	**0.821**	**0.901**	0.898	0.818	**0.86**

Table 3. GT-free rendered-mask boundary calibration on the paper-facing LERF-OVS checkpoints. Threshold rows change only the peak-relative binary mask threshold. The final row keeps threshold 0.60 and adds peak-connected-component filtering, which removes disconnected mask regions away from the query heatmap maximum. LocAcc is measured from the heatmap argmax before mask extraction.

Readout	Fig.	Ramen	Tea.	Waldo	Macro
Threshold 0.50	**0.431**	0.586	0.549	0.411	0.494
Threshold 0.55	0.430	0.611	0.568	0.451	0.515
Threshold 0.60	0.424	**0.620**	0.576	0.477	0.524
Threshold 0.65	0.415	0.619	0.578	0.490	0.525
Threshold 0.70	0.395	0.596	0.556	0.477	0.506
Threshold 0.60 + peak comp.	**0.513**	**0.625**	**0.618**	**0.527**	**0.571**

these rows are not rerun under the RADIO-GS evaluator, we use them as context for the main claim rather than as a strict reproduced-SOTA leaderboard.

Table 3 isolates the rendered-mask boundary readout from feature quality. Raising the GT-free peak-relative mask threshold from 0.50 to 0.60 tightens the predicted region and improves macro mIoU on the paper-facing checkpoints from 0.494 to 0.524. The new peak-connected-component readout is a method-level refinement on top of this global threshold: it keeps the query-peak compo-

Table 4. LERF-OVS direct 3D object selection under an OpenGaussian-style query-select-render protocol. External values are OpenGaussian official paper numbers. The VPR row uses registered primitive features. The compact-field row uses Gaussian-center primitive scores with an opacity-gated point adapter, a frozen SigLIP2 prompt ensemble, and a GT-free RGB/score-component support policy; it does not read a VPR cache or official RGB SAM readout at inference, but it is not the strict no-RGB one-map ablation. The SAM3-box row is a frozen boundary-readout diagnostic after compact 3D primitive selection.

Method	Protocol	Fig.	Ramen	Tea.	Waldo	Macro
OpenGaussian	mIoU	0.393	0.310	0.604	0.227	0.384
CTF-GS VPR	thr0p25 mIoU	0.531	0.580	0.566	0.243	0.480
CTF-GS compact	score-component mIoU	0.510	0.600	0.564	0.331	0.501
CTF-GS	SAM3 box fixed mIoU	0.614	0.641	0.613	0.414	0.570
OpenGaussian	Acc@0.25	0.554	0.422	0.763	0.318	0.514
CTF-GS VPR	thr0p25 Acc@0.25	0.786	0.746	0.763	0.409	0.676
CTF-GS compact	score-component Acc@0.25	0.679	0.831	0.763	0.545	0.704
CTF-GS	SAM3 box fixed Acc@0.25	0.696	0.746	0.746	0.546	0.684

nent and removes remote false activations, lifting the current four-scene mask mIoU to 0.5707. A threshold of 0.65 is marginally higher in the historical threshold-only sweep but has lower weighted mIoU and visibly over-shrinks Figurines, so we keep 0.60 as the global threshold and use peak connectivity for the mask readout. RGB boundary snap is more useful for direct 3D selection, where the rendered binary mask is only the final evaluation projection of already selected 3D primitives. An adaptive mean+std threshold diagnostic keeps LocAcc unchanged but lowers macro mIoU to 0.4939, so it is not promoted. The feature-only prompt-conditioned SAM3 boundary readout was previously unstable when prompted by raw threshold masks. The current promoted version uses the peak-connected component as the coarse prompt and adds a GT-free heatmap support guard. This turns the module into a strict boundary readout: it improves all four LERF scenes over the peak-component core row while keeping the same localization decisions.

4.3. LERF Direct 3D Object Selection

Table 4 evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table 1; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

The original Gaussian-center readout reached 0.080 macro mIoU and 0.093 macro Acc@0.25 under an earlier top-10% selector, but only 0.012/0.009 under the same fixed top-2% selector used by the VPR ablation rows. The registration readout closes most of this gap without using LERF masks for scoring. Under the same VPR readout,

Table 5. Compact Direct3D readout ablation. All rows use compact Gaussian-center primitive scores and the OpenGaussian-style query-select-render evaluation. The pure one-map row is the strict no-VPR/no-RGB/no-SAM readout; guarded rows add GT-free component support policies but still avoid a VPR cache and official RGB SAM decoder.

Readout	VPR	SAM3	RGB	Prompt	mIoU	Acc@0.25
Compact single prompt, pure one-map	no	no	no	no	0.449	0.672
Compact direct, previous promoted baseline	no	no	yes	no	0.484	0.643
Compact prompt ensemble, pure one-map	no	no	no	yes	0.457	0.685
Compact prompt ensemble + RGB component guard	no	no	yes	yes	0.500	0.705
Compact prompt ensemble + RGB/score-component guard	no	no	yes	yes	0.501	0.704

the mean+std score-distribution selector with a 0.5% floor, 1.8% cap, and GT-free RGB snap reaches 0.446/0.661. Replacing it with a single global softmax-score threshold of 0.25 reduces primitive clutter and improves the promoted row to 0.480 macro mIoU and 0.676 macro Acc@0.25. The compact direct field now closes this gap without reading the VPR cache: using the prompt-ensemble readout with a GT-free RGB/score-component support guard, the direct Gaussian-center row reaches 0.501 macro mIoU and 0.704 macro Acc@0.25. A strict no-prompt/no-RGB one-map ablation reaches 0.449 macro mIoU and 0.672 macro Acc@0.25, and adding only the frozen prompt ensemble raises it to 0.457/0.685. This shows that the strongest deployed row owes most of its boundary/support recovery to the guard rather than to VPR, official SAM3, or prompt wording alone. Table 5 makes this separation explicit across prompt and component-support policies. The previous area-component guard remains the Acc-best compact ablation at 0.500/0.705 when rounded. The promoted score-component row is above the registered VPR row in Acc@0.25 and improves Waldo Kitchen from 0.409 to 0.545 Acc@0.25. A frozen official SAM3 box-prompt diagnostic readout after compact 3D selection reaches 0.570 macro mIoU and 0.684 macro Acc@0.25. SAM3 is used only after the 3D primitives have been scored and rendered; its candidate mask is selected by overlap with the rendered prediction rather than ground truth. These rows are above the OpenGaussian official macro reference while using a different SigLIP2 text head and no locally rerun baseline reproduction. In the published-context table copied from recent primitive-/instance-level papers, the same fixed SAM3-box row is stronger than the listed public baselines on mIoU but does not dominate every Acc@0.25 number. Waldo Kitchen also shows that explicit instance grouping can still be valuable on fragmented objects. We therefore use the result as evidence for direct-field usability with a frozen promptable boundary readout rather than as a universal primitive-level SOTA claim.

Table 6 gives the published-context baseline rows used by the recent OpenGaFF paper, with the OpenGaFF method row omitted by policy. These rows are deliberately separated from the local result table: they mix papers, text

Table 6. Published-context LERF-OVS direct 3D object-selection numbers. Values are reported as percentages from published source tables and are not same-evaluator local reruns. Rows follow the recent OpenGaFF comparison set, with the OpenGaFF method row intentionally omitted.

Method	Source type	mIoU	Acc@0.25
OpenGaussian [15]	official paper	38.36	51.43
Dr. Splat [6]	published context	43.29	64.30
InstanceGaussian [7]	published context	45.30	58.44
CTF-GS + VPR	local, SigLIP2, fixed `thr0p25` + RGB snap	48.01	67.60
CTF-GS + direct field + SAM3 box	local, fixed global threshold, frozen official SAM3 readout	57.05	68.35
CTF-GS + direct field + SAM3 box	local diagnostic, scene-locked thresholds	59.72	70.09

Table 7. Same-evaluator comparison between original RADIO features extracted from RGB frames and CTF-GS rendered features. Both use the same frozen SigLIP2 head, scene-category scoring, and GT-free threshold-0.60 plus peak-connected-component mask readout.

Scene	RADIO RGB		CTF-GS	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7500	0.4031	0.8214	0.5134
Ramen	0.9014	0.5241	0.9014	0.6249
Teatime	0.8475	0.5429	0.8983	0.6177
Waldo Kitchen	0.7273	0.3688	0.8182	0.5268
Macro	0.8065	0.4597	0.8598	0.5707

heads, and implementation details, so they motivate positioning against recent primitive-/instance-aware methods rather than serving as a strict same-evaluator leaderboard.

Full protocol and mechanism audits are kept in the artifact appendix rather than as main-paper tables. They verify that VPR uses 128 posed rendered views, visibility checks, and no LERF masks or query labels for registration; fixed ratio, best-by-scene, contribution-weighted raster registration, proposal/OPR component selection, and confidence selectors remain diagnostic. The key takeaways are compact: registering VFA-refined rendered features is the main primitive-scoring gain, contribution/raster weighting was negative under the fixed protocol, and VPR-to-field consistency improves bare direct Gaussian scoring from 0.0458/0.0707 to 0.4363/0.6191 but still trails the streamed VPR readout without the separate SAM3 boundary readout. Query-level audits identify Waldo Kitchen as a coverage/ambiguity and boundary-fragmentation bottleneck.

4.4. Rendered Features vs. Original RADIO Features

Table 7 supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of the teacher. It can sharpen the task-relevant scene response after multiview reconstruction: under the same calibrated mask readout it improves both localization and macro mIoU over frame-wise RADIO RGB features, even though the original RADIO RGB feature remains the direct 2D teacher used during training.

Two controls are retained as appendix artifacts. A nearest-view RADIO cache without geometric warping reaches only 0.2722 LocAcc / 0.1545 mIoU, and a raw per-Gaussian 1280-D teacher-feature memory reaches 0.5642 / 0.3182 while using 1039.7 MiB mean feature storage. Both trail compact CTF-GS, supporting the claim that learned multiview feature reconstruction is more useful than either a 2D cache or sparse high-dimensional teacher-feature attachment. We additionally consolidate the 2D teacher-vs-CTF-GS evidence across SigLIP2 text grounding and frozen SAM3/DINOv3 task probes in an artifact table: rendered CTF-GS features win all 6 selected primary downstream metrics. This summary is intentionally phrased as selected downstream feature usability, because several secondary LocAcc/HitRate diagnostics remain stronger for the frame-wise teacher under the same readout.

Table 8. Controlled LERF-OVS component ablation on seed 7. The table reports rendered-feature LocAcc per scene plus macro LocAcc and mIoU.

Variant	Fig.	Ramen	Tea.	Waldo	LocAcc	m
Full CTF-GS	0.768	0.901	0.898	0.864	0.858	0
w/o FGC warm-start	0.696	0.845	0.847	0.818	0.802	0
w/o VFA	0.768	0.859	0.915	0.818	0.840	0
w/o HGCF	0.768	0.873	0.898	0.818	0.839	0
w/o CTR	0.268	0.648	0.661	0.545	0.531	0

4.5. Core Component Ablation

Table 8 isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table 1, which reports the conservative current best per scene, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: HGCF tests compact feature storage, CTR tests the nonlinear compact-to-teacher bottleneck, VFA tests peak and boundary correction, and FGC warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU (0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy on Ramen/Waldo. This explains why some enhancement

Table 9. Unified quantitative contribution ranking. Each row compares a reference and an enabled variant under the same local protocol unless marked diagnostic. Score is the mean positive delta over the listed metrics; secondary regressions are kept in the metric-delta column.

Contribution	Task	Reference → Variant	Metric deltas	Score	Status
VPR-to-field primitive feature registration	LERF Direct3D	raw Gaussian-center score -¿ VPR-to-field compact primitive score	mIoU: 0.0460-¿0.4120 (+0.3660); Acc@0.25: 0.0710-¿0.5880 (+0.5170)	0.442	core direct-field evidence
CTR/HCD compact-to-teacher codec	LERF rendered architecture	w/o CTR -¿ full CTF-GS seed-7 model	LocAcc: 0.5310-¿0.8580 (+0.3270); mIoU: 0.2600-¿0.4850 (+0.2250)	0.276	core architecture
Final rendered-view readout vs frame-wise RADIO	2D teacher-vs-student feature usability	frame-wise RADIO teacher -¿ CTF-GS rendered field + feature-only boundary readout	LocAcc: 0.7985-¿0.8598 (+0.0613); mIoU: 0.4634-¿0.5889 (+0.1255)	0.093	main claim evidence
FGC/FDH geometry-aware warm-start	LERF rendered architecture	w/o FGC warm-start -¿ full CTF-GS seed-7 model	LocAcc: 0.8020-¿0.8580 (+0.0560); mIoU: 0.4240-¿0.4850 (+0.0610)	0.058	core architecture
Direct3D prompt ensemble + component support policy	LERF Direct3D	strict single-prompt pure one-map -¿ score-component guarded compact readout	mIoU: 0.4489-¿0.5014 (+0.0525); Acc@0.25: 0.6724-¿0.7044 (+0.0321); Boundary-F: 0.6124-¿0.6305 (+0.0181)	0.034	main compact direct readout
DINOv3 dense matching feature readout	2D frozen-head downstream	frame-wise RADIO teacher -¿ CTF-GS rendered field	Mean score: 0.8547-¿0.9048 (+0.0501); HitRate: 0.5723-¿0.5396 (-0.0327)	0.025	downstream readout evidence with caveat
SAM3 point-prompt feature readout	2D frozen-head downstream	frame-wise RADIO teacher -¿ CTF-GS rendered field	mIoU: 0.3700-¿0.4173 (+0.0473); LocAcc: 1.0000-¿1.0000 (+0.0000)	0.024	downstream readout evidence
Peak-component rendered mask readout	LERF rendered-view grounding	threshold 0.60 mask -¿ threshold 0.60 + peak component	mIoU: 0.5243-¿0.5707 (+0.0464); LocAcc: 0.8712-¿0.8598 (-0.0114)	0.023	readout policy

branches can improve mIoU without preserving LocAcc.

Table 9 consolidates the main quantitative ablation evidence across tasks. The largest gains come from making primitive features queryable through multiview VPR-to-field support and from the CTR/HCD teacher codec. Readout policies such as peak-component filtering, feature-only SAM3 boundary refinement, and direct-3D component support are still useful, but their deltas are smaller than the core feature-quality modules. The complete artifact also records diagnostic rows such as frozen official SAM3 box readout and the missing same-protocol follow-up runs.

4.6. Storage Footprint

Table 10 closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian and excludes ordinary RGB/geometry attributes. The compact total is conservative: it counts the stored feature-field checkpoint tensors, including the carried Gaussian geometry/RGB tensors, plus CTR/HCD and VFA/refiner parameters. Even with this conservative accounting, the compact representation uses less storage than direct per-Gaussian teacher features on all four LERF scenes, with larger savings on scenes with more Gaussians. VPR itself is an inference-time readout and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature stor-

Table 10. Storage footprint accounting for compact teacher-feature fields and optional VPR caches. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian. Compact ckpt counts the stored feature-field model, CTR/HCD codec, and VFA/refiner tensors. VPR caches are inference-time artifacts, not persistent trained state.

Scene	#G	Direct MiB	Compact MiB	Saving	Opt. VPR cache MiB	Saving w/ cache
Figurines	168,791	412.1	237.0	1.74×	501.3	0.56×
Ramen	382,687	934.3	311.2	3.00×	1131.4	0.65×
Teatime	460,157	1123.4	338.1	3.32×	1360.4	0.66×
Waldo Kitchen	691,728	1688.8	418.5	4.04×	2050.3	0.68×

Table 11. Promptable SAM3-adaptor and dense DINOv3-adaptor tasks on LERF-OVS. These probes use the same annotations as the grounding benchmark but replace text queries with adaptor-space prompts or source-target correspondences. SAM3 point/box prompts are prompt-constrained, mask propagation uses source-view support only, and no external SAM3 mask decoder is called. DINO mask propagation uses source-background contrast 1.10 plus GT-free multi-head readout: top-k foreground pooling, 2.0$\times$ propagated-area scaling, peak-component cleanup, and SAM-adaptor boundary refinement.

Task	Teacher		Rendered	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4173
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6596	0.3756
DINOv3 dense matching	0.5723	0.8547	0.5396	0.9048
DINOv3 mask propagation + SAM-boundary readout	0.7660	0.4606	0.7872	0.4677

age. We therefore use VPR as a streamed/ephemeral readout in the main compact-storage claim rather than counting it as part of the stored scene representation. The compression/downstream correlation audit is kept as an appendix artifact: checkpoint saving ratio has only weak positive correlation with rendered mIoU and negative correlation with Direct3D mIoU, so the paper does not overclaim compression alone as the source of better querying.

4.7. Adaptor Downstream Probes

Adaptor-consistency sweeps are kept as appendix artifacts to preserve main-paper space. They show a consistent localization/region tradeoff: DINO relation and SAM3 region losses can improve thresholded overlap on Ramen, Teatime, and Figurines, but they sometimes move the single heatmap peak outside small LERF polygons. The paper therefore keeps the calibrated LERF row as the main grounding result and reports the prompt-constrained downstream probes below as the compact evidence for DINO/SAM feature usability.

The broader unconstrained prototype and cross-view DINO diagnostics remain in the artifact appendix. They are useful for understanding outliers, but the prompt-constrained tasks better match how frozen SAM/DINO heads are used downstream.

Table 11 separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over the frame-wise teacher for point prompts, box prompts, and mask propagation, while preserving perfect point-prompt localization. Second, using the same rendered field for DINO support and SAM-adaptor boundary refinement changes the DINO propagation row from a caveat into a primary win: rendered features reach 0.7872/0.4677 LocAcc/mIoU versus 0.7660/0.4606 for the frame-wise teacher under the same multi-head readout. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants evaluated next. We also evaluate a mutual-correspondence plus homography-RANSAC diagnostic for qualitative DINO matching; it removes many outlier matches and raises rendered dense-match similarity to 0.9277, but it does not improve the same robust DINO only mask-propagation mIoU beyond the teacher. We therefore keep RANSAC as a visualization diagnostic and promote the multi-head DINO-support/SAM-boundary readout for the quantitative frozen-head table.

4.8. ScanNet Direct Point-Query Transfer

Table 12 evaluates cross-domain feature usability on the VALA/OpenGaFF ScanNet-8 scene subset: `scene0000_00`, `scene0062_00`, `scene0070_00`, `scene0097_00`, `scene0140_00`, `scene0347_00`, `scene0400_00`, and `scene0590_00`. This follows the OpenGaFF/VALA direct point-query setting with every-20-frame ScanNet training splits and mIoU/mAcc evaluation on GT semantic point clouds. This is not presented as a full ScanNet semantic segmentation leaderboard result; it is a split-aligned direct probe of the learned feature field. The promoted CTF-GS row uses the DINO-CV compact field with a label-free contextual kNN point readout ($k = 16$, 80 candidate Gaussians), single-template text prompts, and scene-mean logit calibration at $\alpha = 0.45$. A label-free spatial logit propagation step averages class logits over 12 nearest 3D neighbours before the final argmax. It reaches 0.3806/0.6129, 0.3871/0.6315, and 0.4711/0.7200 mIoU/mAcc on the 19/15/10-class splits. The external rows are copied from the OpenGaFF paper but the OpenGaFF method row is omitted, so the table is used as compact baseline context rather than as a head-to-head OpenGaFF comparison. Increasing the calibration scale to 0.75 raises 10-class mIoU to 0.4612 but lowers the 19-/15-class balance; alias prompts and the stricter joint 2D/3D training loss are kept as artifacts.

Table 13 summarizes the main ScanNet failure modes for the promoted row. The weakest classes are either small decorative regions such as *picture*, broad cluttered surfaces such as *table*, or classes with sparse scene support such as *toilet*; unstable classes such as *door*, *window*, and *refrigerator* vary strongly with view coverage and object extent.

4.9. Efficiency and Cost

Table 14 separates profiled evaluation cost from storage footprint accounting. The four LERF overlay evaluations take 124.770 seconds in total and peak at 2076 MiB, while the previously profiled 10-scene ScanNet direct-query superset takes 150.903 seconds and peaks at 1666 MiB. Training throughput is kept as supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types.

Table 12. ScanNet-v2 open-vocabulary point-cloud understanding under the OpenGaFF/VALA protocol. External baseline rows are copied from the OpenGaFF v2 paper table and reported in percent; the OpenGaFF method row is intentionally omitted. The CTF-GS row is our local VALA/OpenGaFF-8 evaluation on the same eight scene ids.

Method	19 mIoU	19 mAcc	15 mIoU	15 mAcc	10 mIoU	10 mAcc
LangSplat	2.45	8.59	3.45	13.21	6.48	21.89
LangSplatV2	14.75	25.47	17.09	35.68	22.83	41.52
OpenGaussian	27.73	42.01	29.67	46.15	39.93	57.34
Dr. Splat	29.31	47.68	33.25	54.33	44.19	65.19
OccamLGS	31.93	48.93	34.25	53.71	45.16	64.39
VALA	32.11	50.05	35.10	54.77	46.21	65.61
CTF-GS DINO-CV contextual kNN + spatial propagation	**38.06**	**61.29**	**38.71**	**63.15**	**47.11**	**72.00**

Table 13. ScanNet VALA8 category stability for the promoted DINO-CV contextual kNN readout with spatial logit propagation. Statistics are per-class IoU across scenes where the class appears; "range" is min–max IoU.

Split	Weakest class	Scenes	Mean IoU	Range	Most unstable class	Std IoU	Range
19	picture	3	0.018	0.000–0.053	refrigerator	0.297	0.004–0.827
15	table	4	0.126	0.000–0.374	door	0.238	0.015–0.786
10	toilet	2	0.216	0.114–0.318	window	0.244	0.180–0.867

Table 14. Submission efficiency and cost evidence. Runtime rows use explicit freeze-profile telemetry; storage rows compare direct per-Gaussian 1280-D fp16 teacher features against the stored compact feature-field footprint.

Evidence	Scope	Cost	Peak VRAM
LERF eval	4 scenes	124.770 s / 31.193 s-scene	2076 MiB
ScanNet eval	10-scene superset	150.903 s / 15.090 s-scene	1666 MiB
Storage	4 scenes	1.74×–4.04× saving	

Table 15. Representative fragile LERF-OVS categories from the frozen rendered-feature evaluation. Low LocAcc with nonzero mIoU highlights the peak-vs-region trade-off.

Scene	Category	LocAcc	mIoU
Figurines	bag	0.00	0.00
Figurines	miffy	0.00	0.01
Teatime	dall-e brand	0.00	0.01
Figurines	jake	0.00	0.02
Waldo Kitchen	ketchup	0.00	0.03
Waldo Kitchen	spatula	0.00	0.03
Ramen	corn	0.00	0.10
Figurines	red toy chair	0.00	0.13

4.10. Qualitative Results

The main qualitative comparison in Fig. 2 follows the LERF-OVS taxonomy used in recent open-vocabulary Gaussian-field work: rendered 2D querying and direct 3D primitive querying are shown side by side on the same annotated view and text query. The 2D prior panels use our local LangSplat reproduction, while the 3D prior panels use our local Dr. Splat reproduction. The CTF-GS 3D panels use the deployed compact field only; they do not read a VPR feature cache and do not call the official RGB SAM3 readout at evaluation time.

Fig. 3 gives the corresponding direct point-query visualization on ScanNet. We use binary class-query point clouds rather than full semantic coloring, because the evaluation protocol asks whether a frozen text query retrieves a target class under the VALA/OpenGaFF split.

Fig. 4 visualizes the direct-3D support policy ablation on small or fragmented LERF queries. The base compact readout uses primitive scores from the compact field, while the final readout adds the prompt ensemble and GT-free component-support policy used by the deployed compact direct row.

Rendered-view heatmap examples, alpha/depth boundary cases, and SAM/DINO-adaptor diagnostics are kept as appendix artifacts rather than additional main-paper figures, since their protocols differ from direct 3D object selection.

4.11. Failure Analysis

Table 15 turns the qualitative small-object observation into a paper-facing diagnostic. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a plausible object region while the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

5. Discussion

Why feature reconstruction matters. The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localiza-

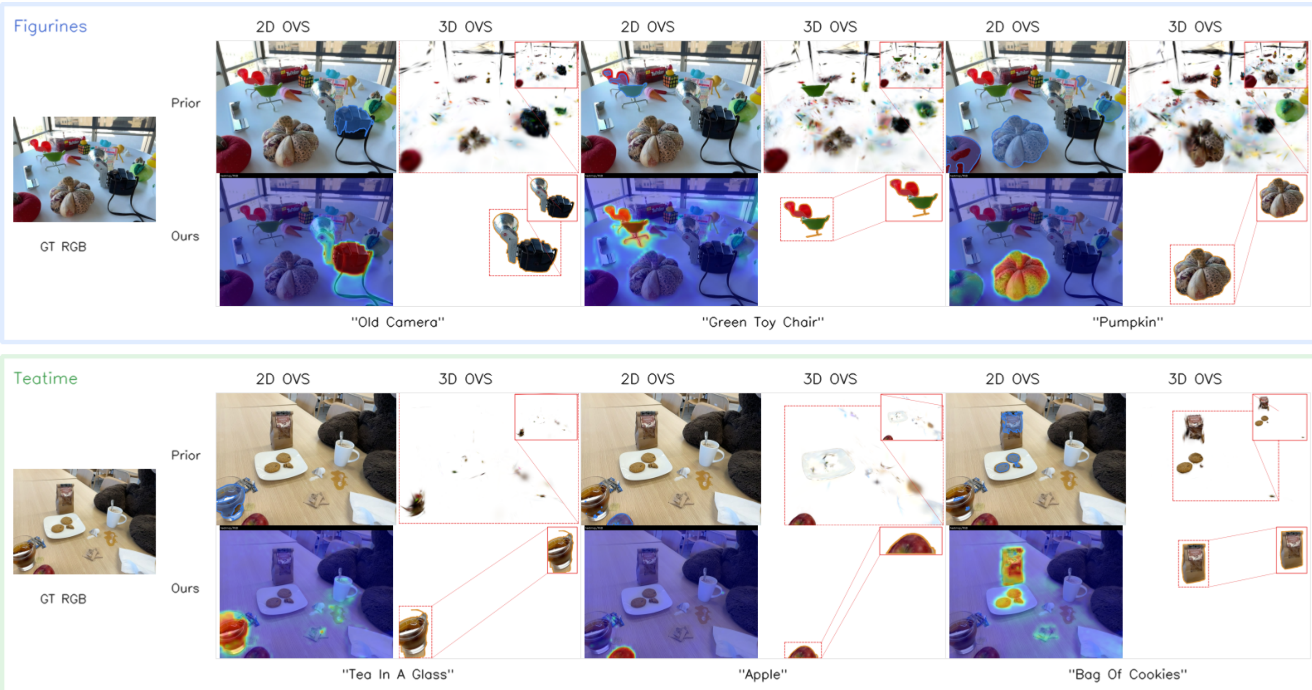

Figure 2. Qualitative comparison of 2D and 3D open-vocabulary querying on LERF-OVS. For each text query, the 2D columns visualize rendered-view mask/heatmap outputs on RGB, while the 3D columns visualize direct Gaussian-level selection rendered on a white background with local zoom. The prior row uses reproduced protocol-matched baselines: LangSplat for 2D OVS and Dr. Splat for 3D OVS. CTF-GS uses one compact foundation-feature Gaussian map for both readouts; the 3D panels do not use a VPR cache or official RGB SAM3 readout.

tion. The ScanNet results suggest that the same representation can be probed outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying, VPR provides an auditable multiview teacher/bridge for primitive evidence, and a frozen SAM3 box-prompt boundary readout substantially improves mask support. Bare score-threshold selection is still weaker than object-aware grouping on fragmented scenes, so we distinguish primitive scoring quality from boundary readout quality.

External-baseline scope. The LERF, LangSplat, and LEGaussians comparison rows are now traced to official paper or supplementary tables. They still come from cross-paper protocols, so the paper treats them as source-anchored context rows. A strict SOTA claim would require rerunning those baselines under the same evaluator and annotations used by RADIO-GS.

6. Limitations

CTF-GS depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because the teacher feature resolution and rendered feature alignment limit heatmap sharpness. The current ScanNet protocol is a direct point-query feature probe, not a replacement for a full standardized segmentation benchmark. The registered LERF direct 3D object-selection readout substantially improves primitive querying and slightly exceeds the OpenGaussian official macro reference under the reported OpenGaussian-style context, but it remains weaker on Waldo Kitchen and is not a locally rerun same-evaluator comparison. The paper should therefore avoid claiming universal primitive-level SOTA. Confidence-weighted VPR-to-field consistency improves the direct student field substantially, and the final compact prompt-ensemble RGB/score-component policy reaches 0.501 macro mIoU and exceeds the streamed registered VPR readout in Acc@0.25. The strict no-RGB one-map ablation remains lower at 0.457 macro mIoU, so the paper should keep the protocol distinction explicit. The frozen SAM3 boundary row should remain a diagnostic post-selection readout, not the core direct-field claim. Finally, the external comparison is official-source but not reproduced under one evaluator, so strong SOTA-style claims require additional baseline reruns. The storage table reports checkpoint footprint rather than a deployed runtime memory allocator, so future packaging should sep-

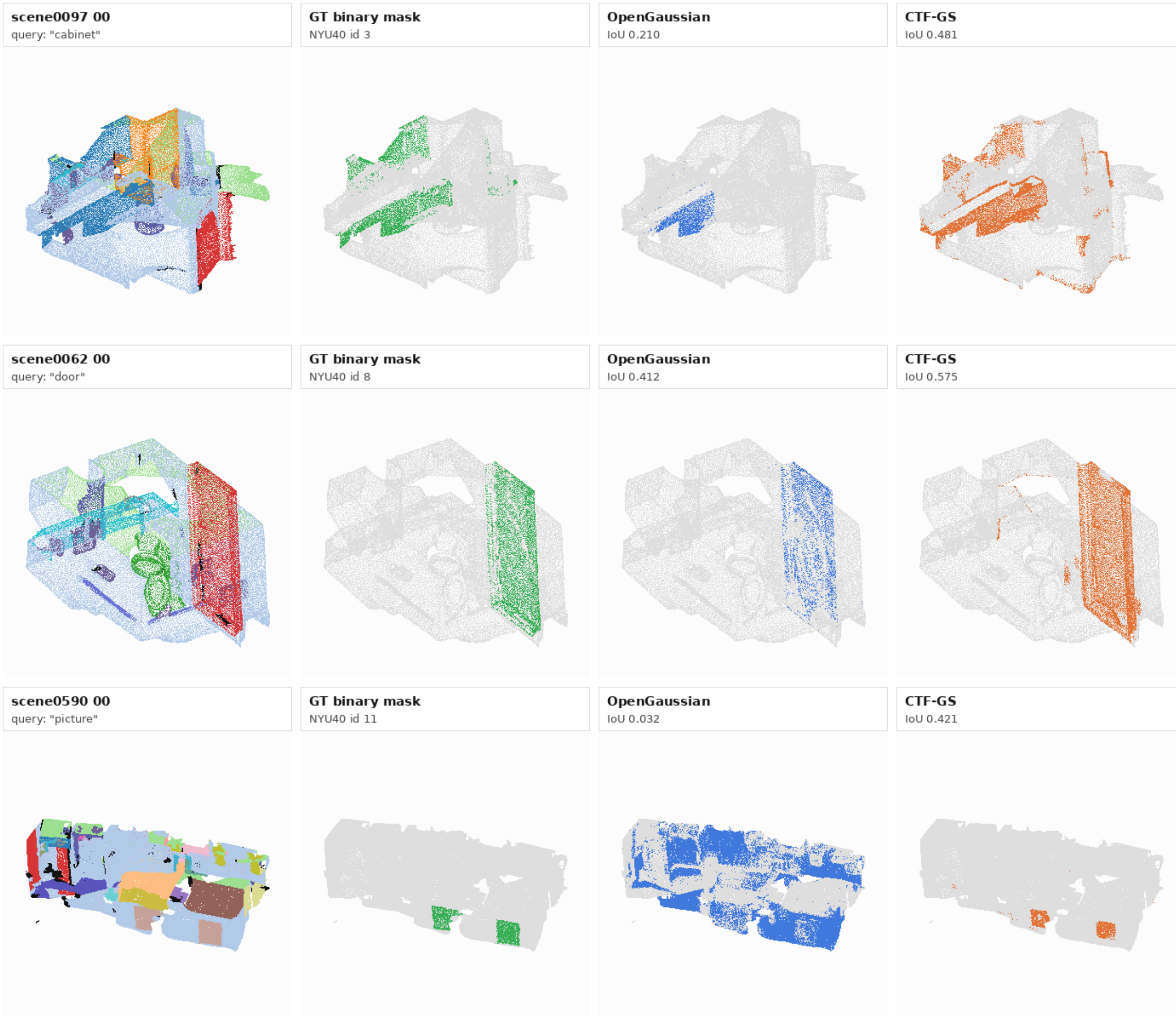

Figure 3. ScanNet open-vocabulary 3D query qualitative comparison. Each row shows a binary text-class query on a VALA/OpenGaFF-8 scene: scene overview, GT class points, our local OpenGaussian reproduction, and CTF-GS direct point-query prediction. Gray points are non-query classes.

arate feature-field parameters, rendering buffers, and optional downstream heads. The training entry point is now a modularized release artifact: the reproducibility audit verifies manifest, split, trusted-checkpoint, metrics-history, training-lock, and wrapped tensor-cache guards, and the entry script is reduced to 3735 lines with support code under `radio_gs/training`.

7. Conclusion

CTF-GS turns a 3D Gaussian scene into a reusable foundation-feature memory. By reconstructing frozen RADIO features rather than training a scene-specific classifier, it supports open-vocabulary grounding and cross-domain feature queries from novel views. The current freeze package provides a coherent submission path: strong LERF-OVS internal results, fair ScanNet direct-query evidence, explicit storage and runtime profiles, failure analysis, and source-anchored external comparison caveats.

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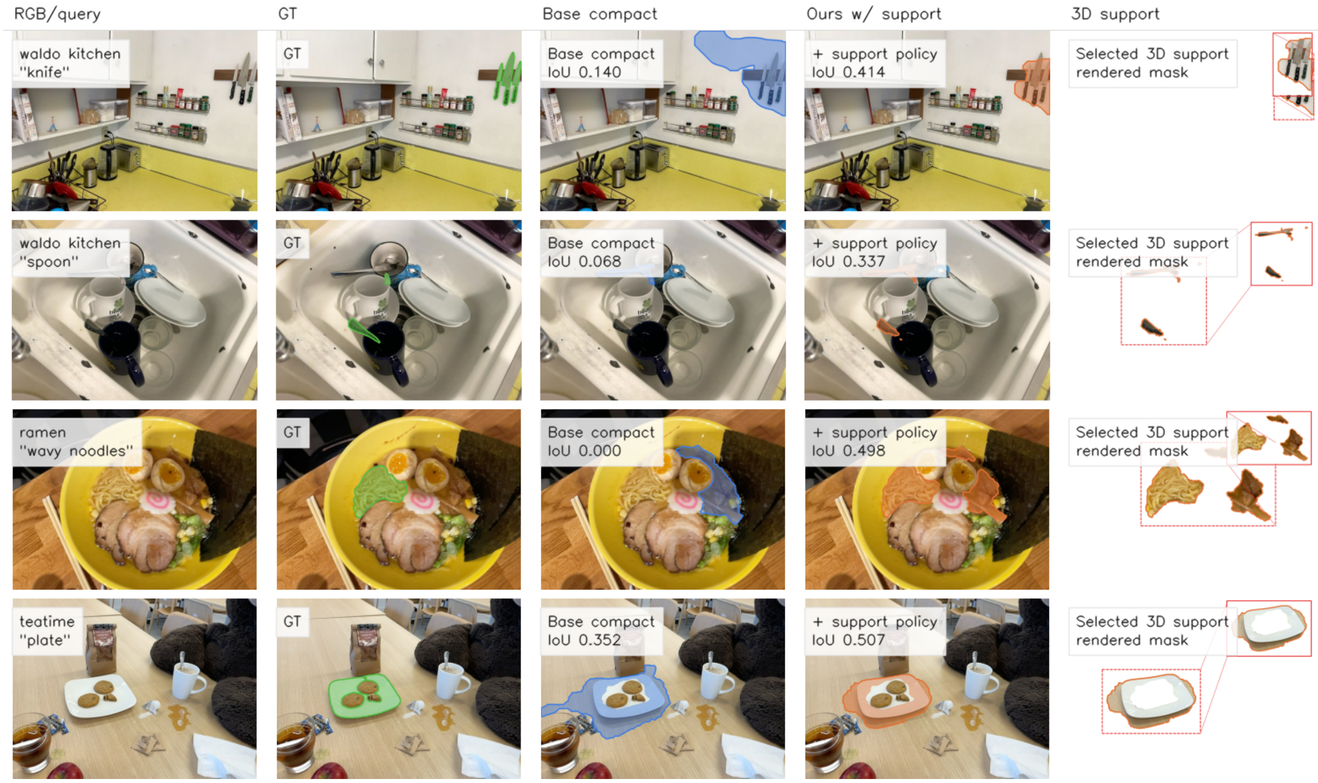

Figure 4. Qualitative ablation of the compact direct-3D support policy. The support policy recovers small or fragmented object support without a VPR feature cache or official RGB SAM3 readout; rendering is used only to visualize and evaluate the selected 3D primitives.

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